MOHAMED AYMAN MOHAMED ABDEL-FATTAH

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate (BSc, Very Good, CGPA 3.1) with practical training in HVAC Design, Mechanical BIM, Plumbing, and Firefighting Systems. Experienced in Revit MEP modeling and coordination workflows, including worksharing, model coordination, documentation, and system modeling, with hands-on application of HAP 4.9 and engineering design calculations. Combining strong technical fundamentals, analytical problem-solving, and a continuous self-learning mindset to deliver accurate, coordinated mechanical solutions.

EXPERIENCE

Pharco Corporation (Engineering & Projects Corp.) **Alexandria, Egypt**

Engineering Excellence Specialist *January 2025 – Present*

- Analyzed continuous facility-wide energy consumption and equipment performance, leveraging technical data to identify optimization opportunities and support corporate conservation initiatives.
- Streamlined critical facility utilities by implementing efficiency modifications for Air Handling Units (AHUs) and managing plumbing networks to reduce overall operational energy usage.
- Executed and documented comprehensive energy-saving projects, tracking methodologies and analytical results to ensure the successful delivery of measurable efficiency improvements.

EDUCATION

Alexandria University **Alexandria, Egypt**

Bachelor of Science in Mechanical Engineering *2019 – 2024*

- Grade: Very Good (CGPA: 3.1)
- Graduation Project: Numerical and Experimental Study on Photovoltaic Thermal Cooling System Using Phase Change Material — Project Grade: A (Excellent).
  - Supervised by: Professor Mohamed Hasab and Associate Professor Mohamed Alnakeeb.
  - Responsible for preparing and modifying the phase change material, contributing to the construction and experimental testing of the complete photovoltaic thermal system.
  - Conducted measurements, data analysis, troubleshooting, and system optimization, ensuring the phase change material met the required specifications.

E.B.S (El-Naser Boys' School) **Alexandria, Egypt**

General Secondary School *Graduated 2019*

- Grade: 390/410 (95.12%)

CERTIFICATIONS & TECHNICAL TRAINING

KAITECH **April 2026 – Present**

Mechanical BIM Diploma [Portfolio / Projects Links]

- Developing practical BIM fundamentals, including BIM workflows, maturity levels, dimensions, ISO 19650, Level of Development (LOD), and BIM roles.
- Training in Revit Mechanical covering HVAC ductwork and piping, plumbing, fire protection, schedules, parameters and annotations, sheets and exports, project management, model federation, project stages, and Revit families.
- Applying Revit worksharing and coordination workflows, including central/local models, worksets, ownership, synchronization, linked models, Copy/Monitor, Coordination Review, and interference checking.
- Developing foundational Dynamo skills covering Dynamo interface and basics, lists, lacing, levels, strings, dictionaries, and mathematical operations.
- Completed Revit Tasks 01–05, including the Families Task, with full scores; additionally completed the Worksharing & Coordination Video (bonus), Worksharing and Coordination Quiz (50/50), and Dynamo Bonus Task 01.

Engineers Syndicate in Alexandria **June 2026 – July 2026**

Firefighting System Diploma [Certificate Link] [Portfolio / Projects Links]

- Completed comprehensive firefighting system design studies covering automatic sprinkler, standpipe & hose, fire alarm, and gaseous fire-suppression systems.
- Performed sprinkler system design and hydraulic calculations, including hazard classification, design area, discharge density, sprinkler flow, pump flow/head, friction losses, and fire-water tank sizing.
- Studied standpipe and hose systems, including system classifications, hose-connection requirements, fire department connections, pressure limitations, piping, valves, and hydraulic calculation procedures.
- Developed knowledge of fire alarm systems, including initiating devices, detectors, notification devices, and conventional, addressable, and wireless fire alarm control panels.
- Studied clean-agent and CO_2 fire-suppression systems, including system components, agent quantity calculations, cylinder selection, nozzle distribution, discharge flow rates, and piping requirements.

Udemy **December 2025 – February 2026**

BIM Revit MEP (Essentials & Advanced) [Course Link]

- Completed two practical Revit MEP projects, building a strong foundation in project setup, modeling, documentation, schedules, and coordination workflows.
- Developed hands-on experience across HVAC, plumbing, drainage, and fire protection systems, including system modeling, sizing, and documentation.
- Gained practical exposure to electrical and ELV modeling, including cable trays, bus ducts, lighting, power circuits, fire alarm, security, data, and CCTV systems.

Udemy **March 2025 – May 2025**

HVAC System Diploma [Certificate Link]

- Developed practical knowledge of HVAC fundamentals, heat transfer, psychrometry, and load types, with application to air-conditioning system design.
- Performed heating and cooling load calculations using HAP 4.9, including space input, examples, and system design reports.
- Studied and applied DX, VRF/VRV, AHU, and chilled-water systems, including equipment selection, AHU sizing, and chilled-water network design.

Engineers Syndicate in Alexandria **December 2024 – January 2025**

Plumbing System Diploma [Certificate Link]

- Completed practical plumbing-system design studies covering domestic water supply, sanitary drainage, storm drainage, swimming pools, and water-heating systems.
- Designed water-supply networks including WSFU-based pipe sizing, flow-rate and pressure calculations, pump selection/head calculations, and water-tank sizing using relevant plumbing codes and design tables.
- Applied sanitary drainage design principles including DFU-based soil & waste pipe sizing, vent-pipe sizing, sump/sewage pits, manholes, traps, and drainage-system components, with reference to the IPC and Egyptian Plumbing Code.
- Developed knowledge of hot-water systems, including storage, instantaneous and central water heaters, hot-water demand calculations, circulation systems, and solar water-heating systems.
- Studied and designed swimming-pool circulation and treatment systems, including pumps, filters, balance tanks, pipe sizing, inlets/outlets, UV treatment, chemical dosing, and pool-heating calculations.

Engineers Syndicate in Alexandria **October 2024 – November 2024**

HVAC System Diploma [Certificate Link]

- Completed an HVAC design project covering cooling-load calculations, system selection, chilled-water system design, pipe sizing, diffuser design, duct sizing, static-pressure calculations, ventilation, and smoke-system design.
- Applied ASHRAE, SMACNA, and NFPA 92 standards within HVAC and smoke-control design.
- Performed cooling-load calculations considering lighting, occupancy, equipment, transmission, solar, and ventilation loads using manual methods and HAP.
- Studied air-conditioning cycles, refrigeration components, and psychrometric-chart applications in HVAC system design.
- Developed knowledge of DX, VRF/VRV, and chilled-water systems, including chilled-water flow-rate, pipe-sizing, and pump-head calculations.

TECHNICAL & INTERPERSONAL SKILLS

- Software (Competent): Revit, AutoCAD, Elite, PipeSizer, HAP 4.9 Hourly Analysis Program, DuctSizer, Daikin Psychrometrics Diagram Viewer, SOLIDWORKS, HTRI, Microsoft Word, Microsoft Excel Microsoft Outlook, Power BI, PowerPoint
- Software (Other): MATLAB, EES, Ansys, COMSOL Multiphysics.
- Interpersonal: Problem Solving, Communication, Teamwork, Self-Motivation, Flexibility, Working Under Pressure, Self-Learning.
- Languages: Arabic (Native), English (Proficient).